

November 20, 2024

Andrew S. Johnston
Executive Secretary
Maryland Public Service Commission
William Donald Schaefer Tower
6 St. Paul Street, 16th Floor
Baltimore, Maryland 21202

Re: RM 87 – Comments of Vehicle-Grid Integration Council on Draft Regulations for Vehicle-to-Grid (“V2G”)

Dear Mr. Johnston,

The Vehicle-Grid Integration Council (“VGIC”)¹ respectfully provides its comments in response to Vehicle-to-Grid (“V2G”) Regulations Proposal filed on October 30, 2024 by the leader of the PC 44 Interconnection Work Group (“Work Group”) to the Maryland Public Service Commission (“Commission”) in Administrative Docket RM 87.

I. Introduction

VGIC is a 501(c)(6) membership-based advocacy group committed to advancing the role of electric vehicles (“EVs”) and vehicle-grid integration (“VGI”) through policy development, education, outreach, and research. VGIC supports the transition to decarbonized transportation and electric sectors by ensuring the value from flexible EV charging and discharging is recognized and compensated to achieve a more reliable, affordable, and efficient electric grid.

II. Comments

VGIC appreciates the work of the PC 44 Work Group in developing the proposed V2G interconnection regulations. As a result of the Distributed Renewable Integration and Vehicle Electrification (DRIVE) Act, Maryland is poised to become a leader in making the transition to EVs affordable, reliable, and clean while also supporting the evolving electric power sector. The draft V2G regulations were developed by PC 44 Work Group contributors, including VGIC, based on best practices across the U.S., and will provide clear, consistent pathways to safely and reliably interconnect both types of bidirectional charging systems: V2G Direct Current (“V2G

¹ VGIC member companies and supporters include American Honda Motor Co., Inc., BorgWarner, Fermata Energy, Ford Motor Company, General Motors, Nissan Group of North America, Bidirectional Energy, Customized Energy Solutions, dcbel, Eaton Corporation, Emporia Corp., EnergyHub, EV.Energy, Flux, Inc., GridBeyond, GridWiz, Hyundai Cradle, Kaluza, Landis + Gyr, Leapfrog Power, Inc., Mercedes-Benz USA, Nuvve Holding Corporation, Orange Charger, Peak Power, PowerFlex, Qcells, Stellantis N.V., Sumitomo Electric, Tesla, The Mobility House, Toyota Motor North America, Inc., WeaveGrid, Clean Power Alliance, Holy Cross Energy, Hoosier Energy, Los Angeles Department of Water & Power, Peninsula Clean Energy, and Sacramento Municipal Utility District. The views expressed in these Comments are those of VGIC, and do not necessarily reflect the views of all individual VGIC member companies or supporters. (<https://www.vgicouncil.org/>)

DC”) and V2G Alternating Current (“V2G AC”). The former is most common today, with available products from BorgWarner, Fermata Energy, Ford Motor Company, Nissan Group of North America, and more. The latter is less common, but emerging, with available products from Tesla and other solution providers.

In developing the draft V2G regulations, the Work Group addressed several important issues related to bidirectional charging systems. First, the draft V2G regulations appropriately define *V2G systems* as “a combination of hardware and software in or around the EVSE and EV for the purposes of communication with and programmed flow of energy into and out of the vehicle battery in support of electrical loads or systems offboard the EV, including the electric grid.”² This definition comprehensively encompasses both V2G DC and V2G AC systems, without being overly-prescriptive as to the use case for which these systems will be enabled.

Second, the draft V2G regulations correctly note that, for the purposes of interconnection, a V2G system is an energy storage device.³ This is consistent with best practices that VGIC has observed, for example in California, where V2G systems, in the context of interconnection, are a type of energy storage system. This avoids the need to entirely “reinvent the wheel” for V2G systems and, instead, leverages the years of experience that utilities, installers, and solution providers have had working with the same underlying safety and reliability standards for inverters (i.e., IEEE 1547-2018 and UL 1741).

Third, the draft V2G regulations establishes a clear pathway for V2G DC systems. The draft also establishes two pathways for V2G AC systems, in recognition that both are capable of ensuring safety and reliability but the market has not yet coalesced around a single pathway.⁴ The Draft Regulation also notes that the two options for V2G AC interconnection are suited for different use cases. Specifically, *Certification pathway 1* can support multiple EVs being used at a certified EVSE, whereas *Certification pathway 2* can support only a single defined EV used as part of a certified AC V2G system, for example, in a single-family home context.

Fourth, the draft V2G regulations set appropriate requirements for V2G systems that operate as V1G Ready EVSE. Specifically, the regulations clarify that V2G Systems operating as V1G Ready EVSE shall be accepted by utilities without an interconnection request and that V1G Ready EVSE shall not be changed to a V2G Ready EVSE without first receiving a permission to operate from the utility. Lastly, clear reporting requirements are set to ensure stakeholders can track the development of V2G systems and make informed recommendations in the PC 44 Work Group or other venues related to DRIVE Act implementation.

Based on these inclusions, the draft V2G regulations are directly aligned with the DRIVE Act, while also supporting the state’s Advanced Clean Cars II policy and Climate Solutions Now Act of 2022. VGIC believes the draft V2G regulations are reasonable and appropriate, and we commend the leader and other contributors to the PC 44 Work Group for diligently addressing the several technical elements needed to ensure safe and reliable V2G interconnection.

² Draft V2G Regulations at 4.

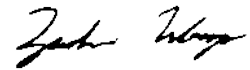
³ *Ibid.*

⁴ Draft V2G Regulations at 5.

With this in mind, VGIC strongly recommends that the Commission approve the draft V2G regulations submitted by the PC 44 Work Group. Establishing these rules is important for expanding customer access to an available VGI technology that will lead to significant benefits, supporting large scale EV adoption and reducing cost impacts to the grid and ratepayers.

III. Conclusion

VGIC appreciates the opportunity to provide comments and looks forward to further collaboration with stakeholders on this important initiative.



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